

FRANCIS NJOGU NJOROGE

BSc. Mechatronic Engineering Student

Nyeri, Kenya | 0702 345 851 | franktronixiroko@gmail.com

PROFESSIONAL PROFILE

Third-year BSc. Mechatronic Engineering student at Dedan Kimathi University of Technology with academic and hands-on exposure to mechanical systems, electronics, electrical systems, computer-aided design, and programming. Practical experience includes an Automated Guided Vehicle (AGV) build, conveyor-system modelling, and design of several mechanical mechanisms. Seeking an industrial attachment from January to March 2027 to build practical skills in automation, control, mechanical systems, electronics, and engineering design.

CAREER OBJECTIVE

To secure an industrial attachment from January to March 2027 that allows application of Mechatronic Engineering coursework to real industrial problems, while developing practical skills in automation, mechanical systems, electrical/electronic systems, control, and engineering design.

EDUCATION

Bachelor of Science in Mechatronic Engineering — 3rd Year, 2nd Semester — *Dedan Kimathi University of Technology*
(2024 – Present)
Registration No.: E022-01-1124/2024

KCSE — Grade A- — *Nyahururu Highway Secondary School*

TECHNICAL SKILLS

Design & CAD	AutoCAD, FreeCAD, SolidWorks, Blender; mechanism design
Programming	Python; working knowledge of C/C++; Arduino; MATLAB; Unity
Automation & Control	Basic PLC knowledge; introductory sensors/actuators; basic control concepts
Electrical & Electronics	Circuit design; basic electrical wiring; introductory motor systems
Mechanical & Workshop	Workshop practice; mechanism design; basic hydraulics and pneumatics

ENGINEERING PROJECTS

Automated Guided Vehicle (AGV) — Group Project — 7-Member Team

- Contributed to electronic circuit work and aspects of mechanical design.
- Assembled and integrated mechanical, electrical, and electronic components into the overall system.
- Assisted with troubleshooting and system integration during development.

Conveyor System Design & Animation — Individual Project

- Independently modelled a conveyor system based on an existing mechanism as a design and learning exercise.
- Built the conveyor model in Blender and explored Unity for system animation.
- Implemented the animation using a reference script with AI assistance, with primary focus on the system model and visualization.

Mechanical Mechanism Design — Individual Coursework Projects

- Designed and studied crank-slider, scissor-lift, four-bar linkage, and cam-follower mechanisms.
- Applied mechanism kinematics and motion-transmission principles to engineering design exercises.

VOLUNTEER EXPERIENCE

Primary School Teaching Volunteer

- Volunteered in teaching and supporting primary school students.
- Explained concepts in a simple and understandable manner.
- Developed communication, patience, and interpersonal skills.

CORE STRENGTHS

Problem Solving • Technical Curiosity • Willingness to Learn • Teamwork • Communication • Adaptability • Practical Approach to Engineering Problems.